

The sentence the book is built on

A BCI measures central-nervous-system activity and turns it into an artificial output a person can use. Cognitive computing is a neighbour: priors and policies. It is not another name for the interface.

AUTHOR

Er. Rishabh Aryan · M.Tech (Artificial Intelligence and Data Science) · Department of Computer Science and Engineering, Indian Institute of Information Technology, Bhagalpur (Bihar). rishabh.250201011@iiitbh.ac.in · rishabharyan887@gmail.com

DEFINITIONS THAT DO NOT MOVE

BCI (1.1)

Measures CNS activity; artificial output that replaces, restores, enhances, supplements, improves, or serves research.

Cognitive computing (8.1)

Systems that estimate, emulate, or adapt to attention, memory, decision, learning. Lives at S6 and S9.

Neuromorphic (8.2)

Hardware/algorithms that imitate neural signalling. Implementation of S3–S5. Not a type of BCI.

Cognitive BCI (8.3)

Target variable y is a cognitive state or content, not a motor or linguistic output.

State vs content (8.4)

State = coarse load/vigilance/valence. Content = the item itself. Products advertise content and deliver state.

Usable trace (2.3)

Predicts a variable on a leakage-safe split at a level that changes the user's options.

THREE TAXONOMIES

Invasiveness: non-invasive / semi-invasive / invasive.

Direction: brain→computer / computer→brain / bidirectional.

Purpose: replace, restore, enhance, supplement, improve, research.

ACTIVE / REACTIVE / PASSIVE

Active: user produces a covert act (imagery, attempted speech).

Reactive: world flashes; decoder uses the evoked response (P300, SSVEP).

Passive: state estimate the user did not issue as a command.

CLAIMS THE BOOK WILL NOT CARRY

- The headset reads your thoughts.
- Non-invasive equals equivalent.
- Invasive equals clinically ready.
- Accuracy of 95 percent. (Which split?)
- AI has solved BCI.
- Cognitive computing is BCI.
- Understands / brain-like AI (unpacked).

RUNNING EXAMPLE

Amina — composite person with ALS. A method that needs two hundred fatiguing trials before breakfast is unfinished.

THE LOOP S0–S9

- S0 Neural source (plastic)
- S1 Sensor / analogue front-end
- S2 Digitise / telemetry
- S3 Preprocess (causal filters)
- S4 Features / embedding
- S5 Decoder $f(\theta)$
- S6 Prior / policy (cognitive computing)
- S7 Actuator / stimulus
- S8 Perception / body / supervisor
- S9 Adapt / log / owner

CONTROL MODES

Open-loop analysis is not an interface. Closed-loop: user sees the output in time to correct. Shared control: decoder proposes; software constrains — disclose it. Bidirectional: S7 writes; artefact and consent rise together.

LATENCY

Loop latency = time from the neural event that should change the output to the first perceptible change. Window length usually dominates. Zero-phase filter is illegal in a live loop.

FOUR PARADIGMS THAT WORK

Motor imagery / SMR

Active. Mu/beta ERD. Window 0.5–2 s. Low alphabet. Failures: illiteracy, EMG shortcut, drift.

P300 / ERP

Reactive. Time-locked oddball. Latency = repetitions. Language model at S6 must be credited.

SSVEP / cVEP

Reactive flicker or codes. Fast if vision allows. Gaze confound: beat an eye-tracker or admit it.

Attempted speech

Active, usually implanted. Linguistic alphabet. Inner speech is not attempt. Listener comprehension is the metric.

MATCH USER TO FAMILY

Flicker safe? SSVEP. Attend rares, no flicker? P300. 2–3 classes enough? Imagery. Conversation + surgery accepted? Speech. Residual gaze reliable? Test an eye-tracker first.

TRACES TO FEATURES

$Z = \text{Phi_feat}(\text{Phi_clean}(X))$. Both maps are part of theta. Tune them on the test block and you trained on the test block.

- Reference is an experiment. Blink on the reference becomes everyone's blink.
- Claim no feature above $f_s/2$. Causal filters only online.
- Artefact audit: nuisance decoder on EOG/EMG. If it wins, stop the neural sentence.
- Overlapping windows may deploy. They may not be shuffled into CV.
- PCA finds variance. CSP finds class-conditional power. Fit on training blocks only.
- Imagery $\rightarrow$ spatially filtered log-power. P300 $\rightarrow$ time-locked shape. SSVEP $\rightarrow$ narrow-band lock. Speech $\rightarrow$ short multi-channel sequences.

LEARNING

Regularised LDA is the baseline of record. A network that cannot beat it on a named split has not earned the paragraph. Chance is a floor, not a baseline. Balanced accuracy when rares are rare.

Permitted splits, in order: later blocks; later sessions; held-out users (Definition 7.3: every statistic refit on training users only).

METRICS THAT SURVIVE TUESDAY

- User-facing communication: corrected meaning per hour of her time, including setup and undos (9.1).
- ITR compares two spellers in one room. It is not a life.
- Credit split (9.3): S5 alone / S6 repair / helper / fail.
- Improvement (10.3): pre-registered function after the device stops, with sham or matched dose.
- Pairing delay is a design constraint. A 1.5 s window may miss the synapse.
- Calibration burden (4.6): minutes to a locked θ that still works on later data.

ITR (SIDE CALCULATION ONLY)

$\text{ITR} = (60/T) [\log_2 N + P \log_2 P + (1-P) \log_2((1-P)/(N-1))]$ bits/min. Zero at chance. T is decision time, not the day.

COGNITIVE COMPUTING — ALLOWED SEATS

Language-model prior over next character. Accumulator that stops P300 flashes. Fade assistance in therapy. Request recalibration. Refuse to emit a workplace score the contract did not name. Not: “the decoder thinks.”

BEYOND THE CLINIC

Wearer-directed (11.1) vs other-directed (11.2). Medical claim (11.3) vs wellness claim (11.4). Neural evaluation (11.5) — judging people with a decoded score — is unfinished as a product: state is not content, and consent under hierarchy is not consent.

ETHICS AS DESIGN

- Consent is a process under power, not a signature.
- Neural data (12.3): minimum collect; frozen purpose; deletion path.
- Abandoned hardware (3.4): name the year-five owner before first surgery.
- Access-complete list (12.4): who cannot use this, by language, cost, vision, caregiver time.
- Intended use (12.2) is one sentence. Changing it is a new device.

REPORTING BUNDLE (12.1)

User · purpose · paradigm · modality · S0–S9 · split · chance · baseline · artefact audit · calibration · credit split · user metric · harms · owner after the study. A 94% attached to an empty bundle is not a result.

LEXICON

Cognitive computing

Neuromorphic

Cognitive BCI

Reads thoughts

Understands

Real-time

Foundation model

priors and policies

hardware style

y is cognitive

forbidden

forbidden for the decoder

name the window first

still needs Tuesday

DESIGN SHEET (MINIMUM)

- Name the user, then the algorithm.
- Name the family, then the filter.
- Name the split, then the score.
- Name S6, then credit S5.
- Name the owner, then the implant.
- Name state or content, then the poster title.

THE THRILL THAT LASTS

A clean definition, a leakage-safe experiment, a user who can still refuse, and a machine that does one modest thing when a nervous system cannot.